Attention from Scratch

Module 5 · Zero to LLM

"It was too big"

The trophy didn't fit in the suitcase because **it** was too *big* / *small*.
Meaning depends on the other words.

Why recurrent nets struggled

1. **Bottleneck** — one fixed-size summary (forward problem)
2. **Vanishing gradients** — training signal fades backwards (different problem!)
3. **Sequential** — no GPU parallelism

The group chat

	asks	example
Query	what am I looking for?	"it" wants a big noun
Key	what do I offer?	"trophy: object, big"
Value	what I pass on	trophy's meaning

One formula

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

dot → scale → softmax → weighted mix

Causal mask

Future positions → `-inf` → weight 0.

Lower-triangular attention.

Multi-head

768 = 12 heads × 64 · each head asks a different question

GPT-2 names: `c_attn` (QKV, with bias) → heads → `c_proj`

Cost

Scores are **n × n** → 2× context = 4× attention work

FlashAttention = same maths, smarter memory

Next: M06 — assemble the GPT block